

Working with the new DNA

UMA 3 presents an improved DNA method designed for easier setup, more features, and simpler execution and debugging. The new system uses structured, asset-driven DNA values to define, group, edit, and apply character shape and coloring through DNA assets and DNAEffects.

DNAGroup

The DNAGroup is a holder for a specific area of DNA – like “Face”, “Body”, etc. The race contains a collection of DNAGroup objects. DNAGroups can contain as many DNA objects that you want, and are created from the context menu Create/UMA/DNA/DNA Group.

DNA

DNA refers to an individual unit of change as presented to the end user within the UMA 3 system. DNA is contained by a DNAGroup, which specifies the category for the DNA. Each DNA entity is uniquely identified by an identifier and is typically initialized with a default value of 0.5. The purpose of DNA is to encapsulate all modifications that are specific to that particular visual change.

The specific changes governed by a DNA are established by associating one or more DNAEffect objects with it. These DNAEffect objects serve as the mechanism through which the DNA exerts influence over the affected objects. Each effect receives an incoming value within the range of 0 to 1 and maps this value to the necessary parameters required to alter the designated object.

DNAEffects

A DNAEffect represents an atomic modification applied to a single target within the UMA 3 system. Each DNAEffect is designed to influence only one specific aspect or element of a character. If you intend to make changes to multiple elements, you must define separate DNAEffects for each target. This ensures that modifications are precisely controlled and organized. The UMA 3 system provides several types of DNA Effects, each tailored to a particular kind of transformation or adjustment. **DNA Effects are cumulative** – so if you set a bone pose, and then modify a bone, and then modify it again, all the effects will add up. So therefore, it's important to understand the order. All Bone poses effects are applied first, during the “PreApply” step to the unmodified rig. Bone transform effects are applied

during the standard Apply step after the rig has been created and modified by the bonepose, and all effects on the DNA are applied in order. You should create any new DNA with this in mind. Blendshapes are applied in the PostApply step, after the mesh and rig have been created.

How value mapping works: Don't worry too much about the mapping - generally, you don't have to do anything if you deal directly with 0-1 values, and you use the default -1 to 1 min/max mapping. Simply type the amount you want to change into the field. If you set a value between 0 and 0.5, it will be converted to a range from 0 down to -1. If you set a value between 0.5 and 1, it will be converted from 0 up to 1. The middle value, 0.5, means no change.

Sometimes, however, you need greater control. In such cases, you can use a curve to create a non-linear mapping. There are many examples of this in the example DNA. When using a curve, 0.5 represents the neutral position—place a key at 0.5, then modify the start and end points as needed to achieve the desired effect.

BoneTranslate: The BoneTranslate Effect applies a positional offset to a chosen bone.

BoneRotate: The BoneRotate Effect applies a rotational offset (specified in degrees) around the specified axis/axes.

BoneScale: Scales along each axis by the amount specified. Note: 0 in an axis specifies not to scale. And amount specified that is not zero is the additional amount to scale – so 0.3 would scale something up by 30%.

BoneTransform: A combine transform adjustment – fades from the current transform to the specified transform. This is a special use case, and may not work exactly like you expect due to interaction with other effects.

BlendShape: Morphs the character along the specified Blendshape. This works directly on the Blendshape on the SkinnedMeshRenderer.

BonePose: This modifies the base skeleton using the specified bone pose.

MeshModifier: This effect applies a MeshModifier. This is used to modify individual vertexes for specific slots. This documentation does not include the documentation for the MeshModifier system.

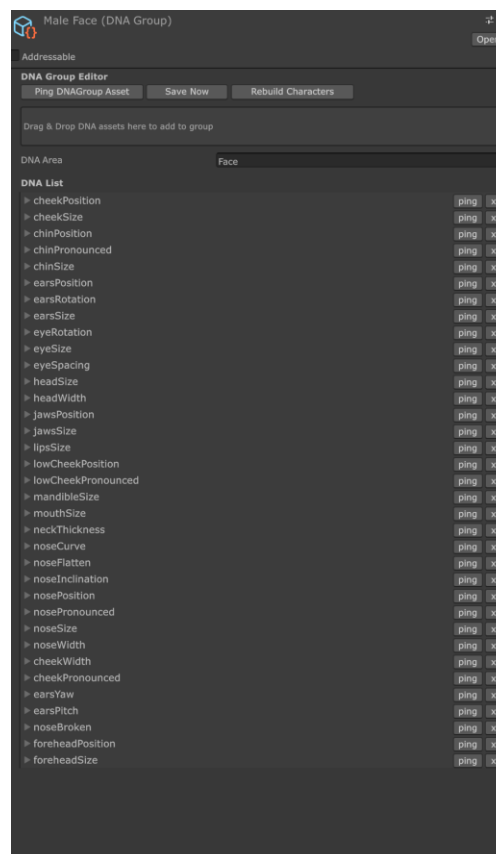
OverlayUVTransform: This effect sets a UV Transform on an overlay. You can use this to move or rotate overlays. UV Tranforms can only apply to Overlay layers. Base overlays are not affected by them.

SharedColor: This effect is used to set a shared color to a lerped value. This effects only one channel of the OverlayColor. Usually, this will be a standard lerp from the “From Color” to the “To Color” using the “Range” method. However, there are different color combination modes you can use (Additive, subtractive, Multiply, Replace). You can specify to use either the Multiply channel (use to colorize) or the Additive channel (use to brighten the color).

SharedColorChannel: This is used to control one individual color component of a specific shared color – for example, you can modify the alpha value of the current shared color. Specify the texture number, the color component (RGBA), and which of the color pairs to modify (Multiply or Additive).

Creating new DNA

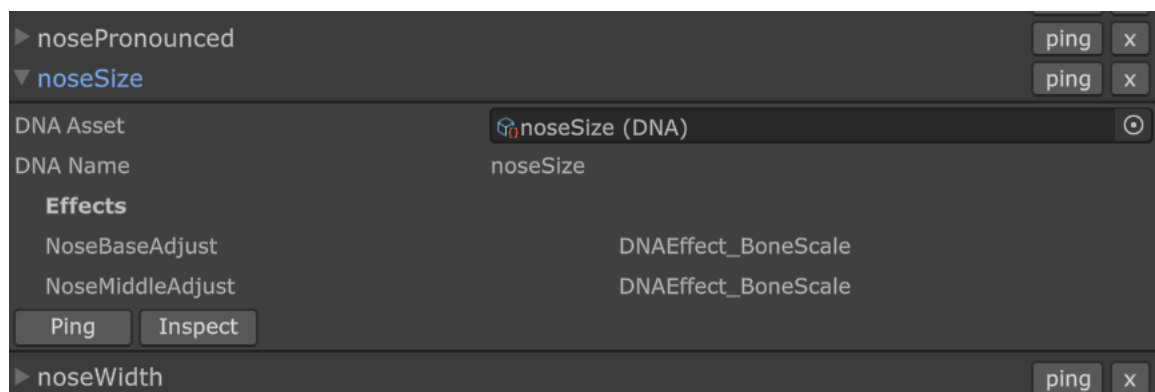
First, create a DNAGroup. Right click in the project window, in the folder you want the DNAGroup to appear, and choose Create/UMA/DNA/DNA Group. Name the DNAGroup something meaningful, and enter a DNA Area (like “Face”, etc.) and we are ready.



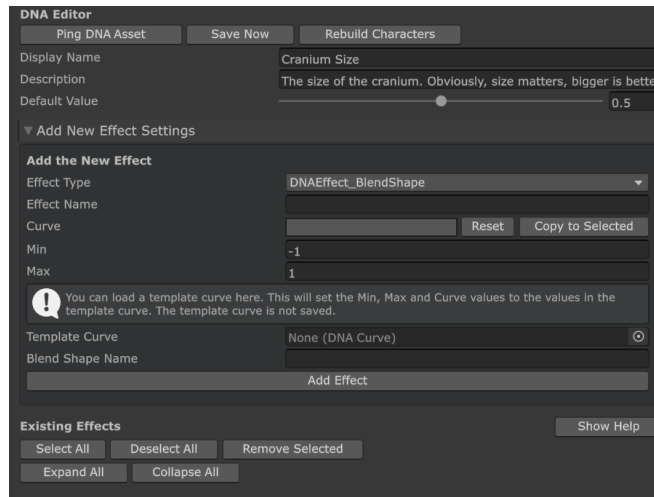
DNAGroup Editor

Next, create the DNA itself by right-clicking in the project and choosing Create/UMA/DNA/DNA Item, and give it a meaningful name. Once the DNA is created, drag and drop it onto the DNAGroup at the location specified. You can see all the DNA you have added to the group, and you can ping the item in the project window, or remove it from the group by pressing the X button.

If you open the DNA by pressing the triangle on the left side, you can see the effects in the DNA, as well as having access to an inspect button to pull it up in the inspector:



To add effects to the DNA, open it in the inspector, and view the DNA. Adding a “Display Name” is optional, but you can use this to ensure that different DNA for different races are presented in a uniform value to the end user (for example, if you have an “age” DNA, you might have different DNA objects with different DNA effects and values for a human than you would for an Orc, but you can present it to the user as “age”, and not have to have a different UI.



DNA Editor

Add effects to the DNA by choosing the Effect Type, giving the effect a name, and the pressing “Add Effect”. Each effect type has a slightly different UI with controls specific to that type. To test your DNA, ensure your group has been added to the race for your character, and then go to the inspector for the character. The checkbox to the left turns the DNA on and off (normally you want this on). But you can enable/disable DNA here to see how DNA works by itself, and in conjunction with other DNA.

Customization

Save Preset

Load Preset

Save AvatarDef

Regen

Editor Time Generation

✓

Default Wardrobe Recipes

Character Colors

Predefined DNA

Assigned New DNA

Reset all DNA

Body

armLength

0.5

Def

Edit

X

armWidth

0.5

Def

Edit

X

belly

0.5

Def

Edit

X

breastSize

0.5

Def

Edit

X

feetSize

0.5

Def

Edit

X

forearmLength

0.5

Def

Edit

X

forearmWidth

0.5

Def

Edit

X

gluteusSize

0.5

Def

Edit

X

handsSize

0.5

Def

Edit

X

height

0.5

Def

Edit

X

legSeparation

0.5

Def

Edit

X

legsSize

0.5

Def

Edit

X

lowerMuscle

0.5

Def

Edit

X

lowerWeight

✓

0.5

Def

Edit

X

upperMuscle

0.5

Def

Edit

X

upperWeight

0.5

Def

Edit

X

waist

0.5

Def

Edit

X

Face

Add New DNA

Group

Body

DNA

bodyFitness

Add DNA Instance

Add All Group DNA

Editing DNA here makes it much simpler to see what is going on. Simply select the edit button to open the DNA is a popup inspector, modify it as needed, and then regenerate the character or drag the slider to view the DNA in action.